When a solution of the bidentate oxalate ligand, $C_2O_4^{2-}$, is added to a solution of $[Fe(H_2O)_6]^{3+}$, the formation of $[Fe(C_2O_4)_3]^{3-}$ occurs in a series of steps that together result in the following overall reaction:

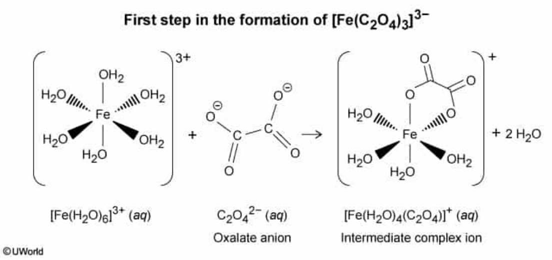
$$[Fe(H_2O)_6]^{3+}(aq) + 3\ C_2O_4^{2-}(aq) \rightarrow [Fe(C_2O_4)_3]^{3-}(aq) + 6\ H_2O(l)$$

During the first step of the reaction, two new coordinate bonds are formed to produce an intermediate ion of:

- A. $[Fe(H_2O)_4(C_2O_4)]^{+}(aq)$. [44%]
- B. $[Fe(H_2O)_2(C_2O_4)_2]^{-}(aq)$. [39%]
- C. $[Fe(C_2O_4)_3]^{3-}(aq)$. [4%]
- D. $[Fe(H_2O)_5(C_2O_4)]^{2-}(aq)$. [11%]

Correct
44% answered correctly

Explanation:



Complex ions (coordination complexes) consist of a central metal ion surrounded by one or more ions or molecules called ligands that are bound to the metal center by coordinate bonds. The coordinate bonds form because the ligands surrounding the metal center act as Lewis bases and donate a lone pair of electrons to the metal center, which acts as a Lewis acid. These Lewis acid-base coordinations hold the complex together, but stronger Lewis bases can displace weaker Lewis bases as ligands within the complex.

In the $[Fe(H_2O)_6]^{3+}$ complex ion, an uncharged water molecule serving as a ligand is a weak Lewis base and can be displaced by the oxalate ($C_2O_4^{2-}$) anion, a stronger Lewis base that can donate **two electron pairs** to the metal atom (ie, oxalate is a **bidentate ligand**). Each $C_2O_4^{2-}$ anion displaces two water molecules and decreases the overall charge of the resulting complex ion by −2. Displacement of water within the complex occurs in a **stepwise sequence**.

In the **first step**, two water molecules are displaced from $[Fe(H_2O)_6]^{3+}$ by a single $C_2O_4^{2-}$ anion to form an **intermediate** complex ion:

$$[Fe(H_2O)_6]^{3+}(aq) + C_2O_4^{2-}(aq) \rightarrow [Fe(H_2O)_4(C_2O_4)]^{+}(aq) + 2\ H_2O(l)$$

In the second and third steps, additional $C_2O_4^{2-}$ anions displace the remaining water ligands to form the final product.

(Choice B) This is the intermediate ion formed during the second step of the reaction, not the first step.

(Choice C) This is not an intermediate ion; it is the end product formed during the final third step of the overall net reaction.

(Choice D) Both the ion charge and this formula are incorrect. The first step adds one $C_2O_4^{2-}$ ion to the complex, but each oxalate ion displaces *two* water molecules and decreases the overall charge of the complex by −2.

Educational objective:
When a reaction occurs in a stepwise sequence, the species formed as products in earlier steps and then subsequently consumed as reactants in later steps are intermediate species. Intermediates do not appear in the overall net reaction.

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